

VOLKEL AP, Netherlands

WMO: 063750

Lat: 51.6586N

Lon: 5.7067E

Elev: 22

StdP: 101.06

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-5.1	-10.5	1.5	-5.5	-8.1	1.9	-4.1	12.8	9.3	11.5	9.3	3.0	70	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	30.0	20.1	27.7	19.4	25.7	18.4	21.2	27.4	20.2	25.9	19.3	24.2	3.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	23.6	18.2	13.2	22.5	17.4	12.5	21.4	61.6	27.5	58.1	25.8	54.9	24.2	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	9.0	7.9	-10.7	34.0	3.3	1.9	-13.1	35.4	-15.0	36.5	-16.9	37.6	-19.3	39.0
			-11.0	23.4	3.3	1.0	-13.3	24.1	-15.2	24.7	-17.0	25.2	-19.4	25.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	3.0	3.7	6.2	9.7	13.4	16.3	18.4	17.9	14.9	11.0	6.7	3.7	(6)
(7)		DBStd	6.55	4.30	4.05	3.51	3.69	3.55	3.22	3.25	2.89	2.80	3.43	3.57	4.31	(7)
(8)		HDD10.0	916	219	178	123	50	9	0	0	0	1	29	108	198	(8)
(9)		HDD18.3	2981	477	411	374	259	158	78	39	44	109	228	348	454	(9)
(10)		CDD10.0	1076	1	1	7	41	113	189	262	244	147	59	10	2	(10)
(11)		CDD18.3	100	0	0	0	1	4	17	42	29	6	1	0	0	(11)
(12)		CDH23.3	1035	0	0	0	16	68	190	423	284	52	3	0	0	(12)
(13)		CDH26.7	299	0	0	0	1	10	49	141	89	9	0	0	0	(13)
(14)	Wind	WSAvg	3.6	4.3	4.2	4.0	3.4	3.3	3.1	3.0	2.9	3.0	3.5	3.7	4.2	(14)
(15)	Precipitation	PrecAvg	799	69	65	52	44	61	69	78	79	64	67	74	78	(15)
(16)		PrecMax	1024	132	138	114	82	114	177	148	166	163	188	125	154	(16)
(17)		PrecMin	607	6	14	6	0	17	15	25	21	7	22	14	25	(17)
(18)		PrecStd	101	33	32	29	20	27	35	36	44	40	31	29	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.5	19.1	25.6	28.3	31.2	33.9	32.4	27.9	22.8	16.8	13.6	(19)
(20)			MCWB	11.1	9.9	12.6	14.8	18.7	20.7	20.8	20.6	19.5	17.5	14.2	11.9	(20)
(21)		2%	DB	11.4	12.0	16.1	21.9	25.2	27.7	30.2	29.4	24.6	19.6	14.6	12.4	(21)
(22)			MCWB	10.0	9.5	11.0	13.9	16.6	19.0	20.4	20.0	18.4	16.0	12.8	11.0	(22)
(23)		5%	DB	10.1	10.7	13.6	19.1	23.1	25.4	27.7	26.6	22.2	17.5	13.2	11.1	(23)
(24)			MCWB	8.7	8.8	10.0	12.7	15.7	18.2	19.6	19.3	17.3	14.9	11.7	9.8	(24)
(25)		10%	DB	8.7	9.4	11.9	16.5	20.7	23.1	25.5	24.1	20.1	16.1	11.9	9.7	(25)
(26)			MCWB	7.5	7.8	9.2	11.6	14.8	17.0	18.7	18.0	16.1	14.1	10.6	8.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.1	13.2	16.3	19.8	22.0	22.9	22.4	20.5	17.8	14.4	12.5	(27)
(28)			MCDWB	12.4	13.0	17.2	23.1	26.1	28.4	29.8	29.6	25.7	21.4	15.7	13.1	(28)
(29)		2%	WB	10.2	10.0	11.7	14.6	18.0	20.2	21.4	20.9	19.0	16.4	13.1	11.2	(29)
(30)			MCDWB	11.2	11.4	14.7	20.2	23.3	25.6	27.7	27.0	23.3	18.8	14.4	12.2	(30)
(31)		5%	WB	8.9	9.1	10.6	13.3	16.5	18.9	20.4	19.8	17.8	15.4	12.0	10.0	(31)
(32)			MCDWB	9.9	10.5	13.0	18.1	21.0	23.8	26.3	25.1	21.2	17.2	13.1	10.9	(32)
(33)		10%	WB	7.7	7.9	9.5	12.2	15.3	17.7	19.3	18.8	16.8	14.3	10.7	8.7	(33)
(34)			MCDWB	8.6	9.4	11.6	15.9	19.7	22.1	24.4	23.1	19.3	15.9	11.7	9.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.2	6.3	8.1	10.2	10.3	10.3	10.2	10.3	9.4	7.7	5.8	4.9	(35)
(36)			MCDBR	5.7	7.4	11.6	14.6	14.3	14.3	14.5	14.6	12.2	9.3	6.4	5.2	(36)
(37)			MCWBR	5.0	5.4	7.4	7.8	7.0	6.8	6.1	6.5	6.6	6.1	5.1	4.6	(37)
(38)		5% WB	MCDBR	5.5	6.4	9.8	12.6	12.1	12.1	12.6	12.8	10.6	8.6	6.1	5.1	(38)
(39)			MCWBR	5.1	5.0	6.6	7.1	6.6	6.4	6.0	6.5	6.2	6.0	5.2	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.358	0.390	0.422	0.428	0.425	0.435	0.425	0.398	0.371	0.347	0.330	(40)	
(41)		taud	2.347	2.316	2.233	2.179	2.207	2.232	2.212	2.260	2.324	2.379	2.356	2.323	(41)	
(42)		Ebn at Noon	662	751	794	812	826	832	815	802	780	727	638	596	(42)	
(43)		Edh at Noon	64	86	112	132	135	133	133	121	102	80	63	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.73	1.41	2.58	4.15	5.01	5.39	5.08	4.32	3.17	1.85	0.86	0.56	(44)	
(45)		RadStd	0.09	0.27	0.36	0.44	0.41	0.42	0.60	0.38	0.30	0.15	0.09	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (90 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page